Sumedh Hindupur | CV

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Education

Harvard University — Cambridge, MA, USA
Ph.D. in Applied Mathematics — *2023–(Expected) 2027*
S.M. in Applied Mathematics, GPA: **4/4** — *2023-2026*
Advisor: Demba Ba

Indian Institute of Technology Bombay — Mumbai, India
B.Tech & M.Tech: Electrical Engineering (Signal Processing), Minor: Physics, GPA: **9.77/10** — *2018– 2023*
Institute Silver Medal & *Bhavesh Gandhi Prize* (Rank 1/78 students)
Thesis: "Model Learning in Partially Observed Markov Decision Processes" (*Exceptional Research Award*)
Thesis Advisor: Vivek Borkar

Research Interests

Neural Representation Geometry, Machine Learning Interpretability, Computational Neuroscience

Experience

- **Research Intern**, Physics of AI Lab, NTT Research, USA — *May 2025-August 2025*
  Mentors: Ekdeep Singh Lubana, Hidenori Tanaka
- **Research Intern**, Hokkaido University, Japan (remote) — *Dec 2020- July 2023*
  Mentors: Hideaki Shimazaki, S. Amin Moosavi

Publications

(† *denotes equal contribution*)

Journal Papers:

1. S. Amin Moosavi†, **Sai Sumedh R. Hindupur**†, and Hideaki Shimazaki. "Population coding under the scale invariance of high-dimensional noise.", *Science Advances* (2026), *DOI:10.1126/sciadv.adz9632*. [paper link]

Conference Papers:

1. Ekdeep Singh Lubana†, Can Rager†, **Sai Sumedh R. Hindupur**†, et al., "Priors in Time: Missing Inductive Biases for Language Model Interpretability", *International Conference on Learning Representations (ICLR)*, 2026. [paper link]
2. **Sai Sumedh R. Hindupur**†, Ekdeep Singh Lubana†, Thomas Fel† and Demba Ba. "Projecting Assumptions: The Duality between Sparse Autoencoders and Concept Geometry", *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. [paper link]
3. **Sai Sumedh R. Hindupur** and Vivek S. Borkar. "Online Parameter Estimation in Partially Observed Markov Decision Processes." 2023 59th Annual Allerton Conference on Communication, Control, and Computing (Allerton). IEEE, 2023. [paper link]

Manuscripts under Review:

1. **Sai Sumedh R. Hindupur**, Hadas Orgad, Thomas Fel and Demba Ba, "When Models Don't Manipulate Manifolds: The Geometry of a Comparison Task", *under review at ICLR 2027*. [paper link]
2. Gaia Grosso, **Sai Sumedh R. Hindupur**, Thomas Fel, Samuel Bright-Thonney, Philip Harris, Demba Ba, "Sparse, self-organizing ensembles of local kernels detect rare statistical anomalies", *under review at PNAS*, *arXiv:2511.03095*, 2025. [paper link]

Workshop Papers:

1. **Sai Sumedh R. Hindupur**, Thomas Fel, and Demba Ba, "On the Dissociation of Order and Identity in Number Manifold Representations", Accepted at *NeurReps: Symmetry and Geometry in Neural Representations, NeurIPS 2026.*

2. David Cao, **Sai Sumedh R. Hindupur**, and Demba Ba, "From States to Dynamics: Detecting Rule Violations in OthelloGPT", Accepted at *Interpretability as a Science Workshop (InterpScience), NeurIPS 2026.*

Invited Talks

- **EE Department Seminar, IIT Bombay** (Mumbai, India): "Geometric Mechanistic Interpretability" *Jan 2026*
- **Caltech NeuroAI Journal Club** (Zoom): "Duality between Sparse Autoencoders & Concept Geometry" *May 2025*

Leadership & Mentorship

Co-Founder, PRISM Reading Club [prism-interp-neuro.github.io] *Fall 2025-Present*
Harvard SEAS and Kempner Institute | Prof. Demba Ba

– Co-founded **PRISM** (Perspectives on Interpretability in Sciences and ML), a weekly reading group systematically exploring the intersection of neuroscience and ML interpretability.
– Organize **panel discussions** and **paper presentations** including ML interpretability and neuroscience researchers; sessions draw graduate students, postdocs, and faculty from SEAS, MCB, IAIFI, and the Kempner Institute.

Research Mentor
Harvard University

– **David Cao** (Undergraduate): Supervising research on how OthelloGPT encodes game rules. *Feb 2026 - Present*
Outcome: Accepted workshop paper at InterpScience Workshop, NeurIPS 2026
– **Terry Zhou** (Masters student, Data Science): Supervising research on weight-space vs activation space interventions in language models. *May 2026 - Present*

Teaching Experience

Harvard University

- AM231/ES201- Decision Theory (Teaching Fellow) *Spring 2026*
- AM220- Geometric Methods in Machine Learning (Teaching Fellow) *Spring 2025*

IIT Bombay

- EE736- Introduction to Stochastic Optimization (Teaching Assistant) *Spring 2023*
- EE659- A first course in Optimization (Teaching Assistant) *Fall 2022*

Honors and Awards

- Received the **Nicole A. Chen and Karina A. Chen Graduate Student Research Fellowship** at Harvard *2025*
- Awarded **Institute Silver Medal** and **Bhavesh Gandhi Memorial Prize** for **ranking first** among 78 students in Electrical Engineering Dual Degree Batch of 2023 at IIT Bombay *2023*
- Recipient of **OP Jindal Engineering** & **Management Scholarship** for academic and leadership excellence *2019*
- Secured All India Rank **170/200k** in **JEE Advanced** and **476/1 million** in **JEE Mains** *2018*

Academic Service

- Invited reviewer for NeurIPS 2026 (main conference) *2026*
- Reviewer for NeurReps and InterpScience Workshops at NeurIPS 2026 *2026*

Selected Research Contributions

The Geometry of a Comparison Task *February 2026 - September 2026*

Research Assistant | Harvard University | Prof. Demba Ba

- Used **representation geometry** and **causal interventions** to translate internal language model computations (Qwen2.5-7B-Instruct) on pairwise number comparisons into **concise algorithms**
- Discovered the use of **linear representations of numbers for computation** by the language model, despite the presence of curved number manifolds
- Demonstrated an extension of the pairwise number comparison algorithm to longer sequences using **three-number comparisons**

Population Coding under Scale Invariance of High-Dimensional Noise *July 2021 - August 2026*

Research Collaboration | Prof. Hideaki Shimazaki | Dr. S Amin Moosavi

- Investigated the **population coding characteristics** of neurons in the primary visual cortex of mice
- Performed **data analysis on large-scale calcium imaging datasets** (>10,000 neurons) to discover **scaling laws** in eigenspectra of neural activity and alignment of signal with noise
- Simulated **synthetic neural populations** to control scaling of information and validate theoretical insights

SparKer: Sparse Local Kernels for Anomaly Detection *Fall 2025*

Research Assistant | Harvard University | Led by Gaia Grosso | Prof. Demba Ba

- **Co-designed SparKer**, comprised of competing Gaussian kernels, to detect statistical anomalies in data
- Developed theoretical insights into **learning dynamics** of SparKer to understand its anomaly detection abilities
- Highlighted **desiderata for anomaly detection methods** across domains (particle physics, language, vision models)

Missing Inductive Biases for Language Model Interpretability *May 2025 - Sep 2025*

Research Intern | Physics of AI Lab, NTT Research | Ekdeep Singh Lubana

- Identified the **temporal priors** of existing Sparse Autoencoders (SAEs) for interpreting language models
- Investigated **non-stationary structure** in language model (LM) representations and its mismatch with SAE priors
- Assisted in developing **Temporal Feature Analyser**, a novel tool for temporal structures present in language

Duality between Sparse Autoencoders and Concept Geometry *Sep 2023 - March 2025*

Research Assistant | Harvard University | Prof. Demba Ba | Collaborators: Ekdeep Singh Lubana, Thomas Fel

- Discovered **implicit geometric assumptions** made by existing SAEs about concept geometry through a novel **receptive field** perspective on SAE encoders, highlighting pitfalls of using proxy metrics to compare SAEs
- Constructed a **novel SAE, SpaDE** (Sparsemax Distance Encoder), to capture nonlinearly separable and heterogeneous concepts that other SAEs systematically fail to capture
- Demonstrated real-world applicability of SAE limitations using vision and (formal) language models

Model Learning in Partially Observed Markov Decision Processes *June 2022 - June 2023*

Dual Degree Project | IIT Bombay | Prof. Vivek S Borkar

- Developed an online algorithm employing **extremum seeking** and stochastic approximation for parameter estimation with partial information in a POMDP, and derived convergence guarantees for the algorithm
- Implemented the algorithm on a gridworld environment and a queueing system